

Percentage

By:-

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PERCENTAGE

- ❑ Percentage is the representation of value expressed as a fraction of 100.
- ❑ Percent means for every hundred.

Important Points

1. a is what % of b or What % of b is a?



$$\frac{a}{b} * 100$$

2. % Change = difference / initial value * 100



$$\frac{a - b}{b} * 100$$
$$\frac{a - b}{a} * 100$$

3. x% of y = y% of x

4. If several percentages are acting on the same number then we can add all the percentages.

$$A\% \text{ of } (K) + B\% \text{ of } (K) + C\% \text{ of } (K) \dots = (A + B + C \dots)\% \text{ of } (K)$$

Important Points

If a number K got increased by A% and B% successively then the final percentage is given by

$$[A + B + (AB/100)] \%$$

Note1: If decreased then substitute +A% with -A%

CONVERTING FRACTION INTO %

❑ To convert fraction into % , multiply the given fraction by 100.

$$\frac{1}{2} = 50\%$$

$$\frac{1}{3} = 33\frac{1}{3}\%$$

$$\frac{1}{4} = 25\%$$

$$\frac{1}{5} = 20\%$$

$$\frac{1}{6} = 16\frac{2}{3}\%$$

$$\frac{1}{7} = 14\frac{2}{7}\%$$

$$\frac{1}{8} = 12\frac{1}{2}\% = 12.5\%$$

$$\frac{1}{9} = 11\frac{1}{9}\%$$

$$\frac{1}{10} = 10\%$$

$$\frac{1}{11} = 9\frac{1}{11}\%$$

$$\frac{1}{15} = 6\frac{2}{3}\%$$

CONVERTING % INTO FRACTION

❑ To convert % into fraction, divide the given % by 100.

$$100\% = 1$$

$$50\% = \frac{1}{2}$$

$$33.33\% = \frac{1}{3}$$

$$25\% = \frac{1}{4}$$

$$20\% = \frac{1}{5}$$

$$16.66\% = \frac{1}{6}$$

$$14.2857\% = \frac{1}{7}$$

$$12.5\% = \frac{1}{8}$$

$$11.11\% = \frac{1}{9}$$

$$10\% = \frac{1}{10}$$

$$33.33\% = \frac{1}{3}$$

$$9.09\% = \frac{1}{11}$$

$$8.33\% = \frac{1}{12}$$

Covert following into fraction

$$106\frac{2}{3}\%$$

$$366\frac{2}{3}\%$$

$$457\frac{1}{7}\%$$

$$106\frac{2}{3}\% = 100\% + 6\frac{2}{3}\%$$

$$= 1 + \frac{1}{15}$$

$$= \frac{16}{15}$$

$$366\frac{2}{3}\% = 300\% + 66\frac{2}{3}\%$$

$$= 3 + 2 \times \frac{1}{3}$$

$$= \frac{11}{3}$$

$$457\frac{1}{7}\% = 400\% + 50\% + 7\frac{1}{7}\%$$

$$= 4 + \frac{1}{2} + \frac{1}{14}$$

$$= \frac{56 + 7 + 1}{14}$$

$$= \frac{64}{14}$$

$$= \frac{32}{7}$$

Find

$$45.45 \% \text{ of } 3333 \Rightarrow 5(9.09 \%) \text{ of } 3333 \Rightarrow 5\left(\frac{1}{11}\right) \times 3333 \Rightarrow 5 \times 303 \Rightarrow 1515$$

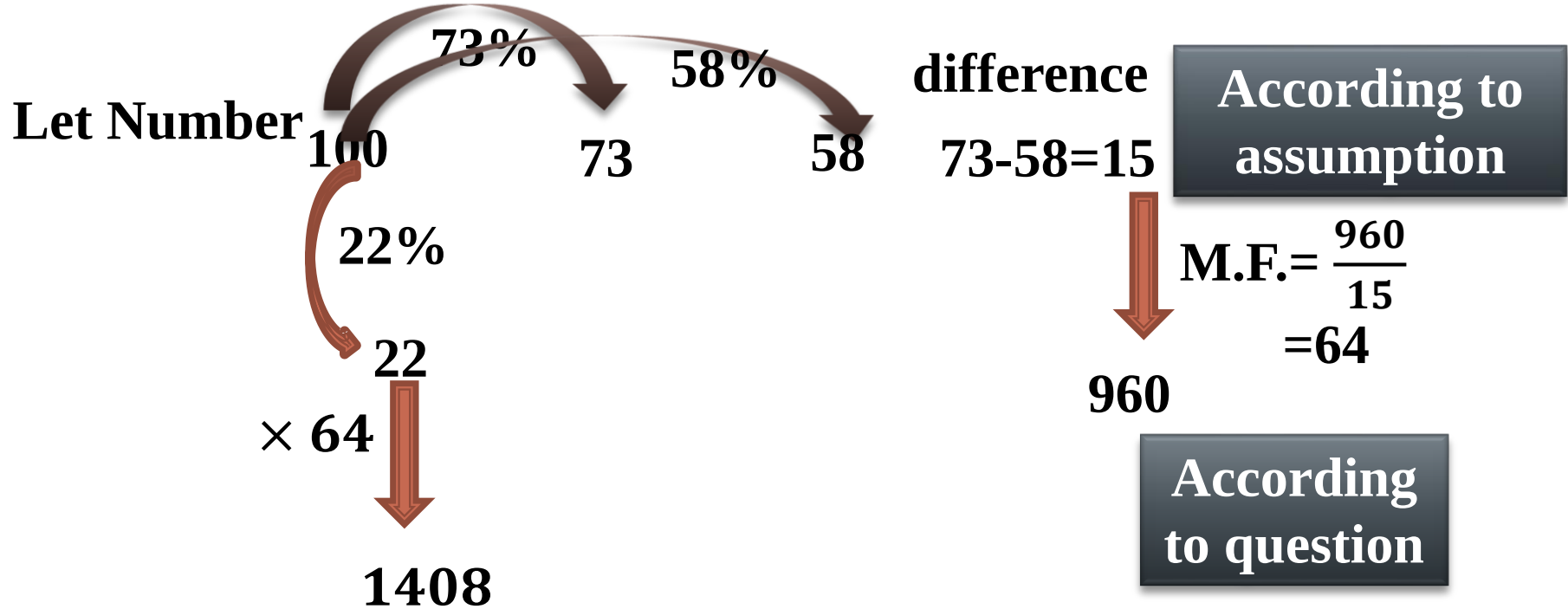
$$54.54 \% \text{ of } 6600 \Rightarrow 6(9.09 \%) \text{ of } 6600 \Rightarrow 6\left(\frac{1}{11}\right) \times 6600 \Rightarrow 6 \times 600 \Rightarrow 3600$$

$$66.66 \% \text{ of } 666 \Rightarrow 6(11.11 \%) \text{ of } 666 \Rightarrow 6\left(\frac{1}{9}\right) \times 666 \Rightarrow \frac{2}{3} \times 666 \Rightarrow 444$$

$$24\% \text{ of } 225 \Rightarrow 225 \% \text{ of } 24 \Rightarrow (200 + 25) \% \text{ of } 24 \Rightarrow 48 + 6 \Rightarrow 54$$

$$45\% \text{ of } 2662 \Rightarrow (50 - 5) \% \text{ of } 2662 \Rightarrow 1331 - 133.1 \Rightarrow 1197.9$$

If difference of 73% of a no and 58% of no is 960,
find 22% of that no.



If a price of sugar is increased by 25% then by how much % the consumption must be reduced so that there is no change in expenditure.

Let price = 100
& quantity = 100

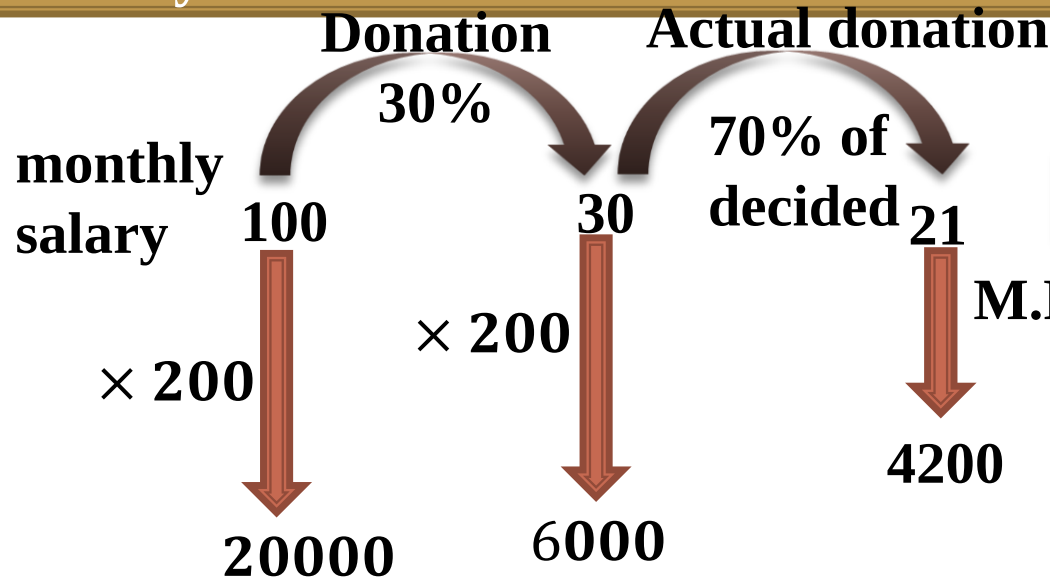
price increased
25%

= 125 New price
= 100

Now no change in expenditure
Means price = 100

$$\begin{aligned}\% \text{ consumption must be reduced} &= \frac{\text{difference}}{\text{Initial value}} \times 100 \\ &= \frac{25}{125} \times 100 = 20\% \text{ decrease}\end{aligned}$$

Aastha donates 30% of her monthly salary on orphanage but on the day of donation, she donates Rs 4200 which is 70% of what she decided. What is the actual donation? What is her actual salary?



According to assumption

$$\text{M.F.} = \frac{4200}{21}$$

$$= 200$$

According to question

If a side of square is increased by 30% find % change in the area of square?

Case 1

Let side =100

$$\begin{aligned}\text{Area} &= 100 \times 100 \\ &= 10000\end{aligned}$$

Case 2

side of square is increased by 30%

New side =130

$$\begin{aligned}\text{New Area} &= 130 \times 130 \\ &= 16900\end{aligned}$$

$$\% \text{ change} = \frac{\text{difference}}{\text{Initial value}} \times 100$$

$$= \frac{6900}{10000} \times 100 = 69\% \text{ increase}$$

If the radius of circle is increased by 25% then find the % change in its area?

Case 1

$$\text{Let } r = 100$$

$$\begin{aligned}\text{Area} &= \pi \times 100 \times 100 \\ &= \pi \times 10000\end{aligned}$$

Case 2

Radius of circle is increased by 25%

$$\text{New radius} = 125$$

$$\begin{aligned}\text{New Area} &= \pi \times 125 \times 125 \\ &= \pi \times 15625\end{aligned}$$

$$\% \text{ change} = \frac{\text{difference}}{\text{Initial value}} \times 100$$

$$= \frac{5625 \pi}{10000 \pi} \times 100 = 56.25\% \text{ increase}$$

If the base radius of right circular cylinder is increased by 40% and its height is reduced by $37\frac{1}{2}\%$ then find the % change in its volume?

We know $40\% = \frac{2}{5}$ & $37\frac{1}{2}\% = \frac{3}{8}$

Case 1

Case 2

Let radius $R=5$

Let height $H=8$

Volume $V = \pi \times 5 \times 5 \times 8$

$$V = 200\pi$$

$$= 7$$

New Volume

$$V' = \pi \times 7 \times 7 \times 5$$

$$= 245\pi$$

$$\% \text{ change} = \frac{\text{difference}}{\text{Initial value}} \times 100$$

$$= \frac{45\pi}{200\pi} \times 100 =$$

22.5% increase

1. (a) 67.5% is equal to
- A. $\frac{27}{40}$ B. $\frac{5}{8}$ C. $\frac{11}{20}$ D. $\frac{7}{8}$ E. $\frac{17}{20}$

1. (b) $9/16$ is equal to
- A. 56.25% B. 68.75% C. 60.75%
- D. 64.25% E. None of these
- 

1. (c) 64% of which number is equal to 432?
A. 575 B. 625 C. 675 D. 725 E. None of these

(d) If 45% of N is equal to 65% of 720, then what is the value of N?
A. 960 B. 1040 C. 1200 D. 1240 E. None of these



2 If 30% of x is equal to 20% of 420, then find x .

A. 280 B. 320 C. 140 D. 840 E. None of these

3 In a test, Mr. John and Mr. Mehta scored 50 marks and 60 marks respectively. By what percent were the marks of Mr. Mehta more than that of Mr. John?

- A. 10% B. 25% C. 20% D. $16\frac{2}{3}\%$ E. None of these**
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4. Two students Sharan and Karan took a test. Sharan got 65% of the maximum marks in the test and Karan got 55% of the maximum marks of the test. The difference between the marks obtained by Sharan and Karan is 48. What are the maximum marks of the test?

A. 360 B. 400 C. 440 D. 480 E. None of these



5 Ananya's monthly income was increased by 15% and became ₹7406.

What was her monthly income before the increase?

A. ₹6840

B. ₹6240

C. ₹6640

D. ₹6440 E. None of these

6.

(a) What percentage of 180 is 54?

A. 70%

B. $333\frac{1}{3}\%$

C. 30%

D. $33\frac{1}{3}\%$

E. None of these

(b) What percentage of 54 is 180?

A. 70%

B. $333\frac{1}{3}\%$

C. 30%

D. $33\frac{1}{3}\%$

E. None of these

6.

(a) What percentage of 180 is 54?

A. 70%

B. $333\frac{1}{3}\%$

C. 30%

D. $33\frac{1}{3}\%$

E. None of these

(b) What percentage of 54 is 180?

A. 70%

B. $333\frac{1}{3}\%$

C. 30%

D. $33\frac{1}{3}\%$

E. None of these

7. If Arun's monthly salary is 25% more than the monthly salary of Bhupathi, then by what percentage is the monthly salary of Bhupathi less than that of Arun ?

A. 10% B. 12% C. 15% D. 20% E. None of these

8. The price of rice is increased by 20% By what percentage should the consumption be reduced so that the expenditure on rice remains the same?

A. $16\frac{2}{3}\%$ B. $66\frac{2}{3}\%$ C. 20% D. 25% E. None of these



9. The price of petrol is first increased by 40%, then it is decreased by 20%. The latest price of a litre of petrol is 56. Then what was the original price of the petrol per litre?

- A. ₹48 B. ₹50 C. ₹62 D. ₹55 E. None of these**
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10. Mr. Naveen Jain donates 14% of his monthly income to charity. This month he gave 1,624 to charity which is 80% of what he donates every month. What is the monthly income of Mr. Naveen Jain ?

- A. ₹12,500 B. ₹16,500 C. ₹14,500**
D. ₹18,500 E. None of these

11. In an exam Gopi got 30% of maximum marks and failed by 35 marks. Bhaskar got 50% of the maximum marks and got 25 marks more than the pass marks. Find the maximum marks.

A. 350 marks B. 400 C. 300 D. 500 E. None of these



- 12. The ratio of the number of male employees and female employees in a company is 5: 2. If 60% of the male employees and 25% of female employees are post-graduates, then what percentage of the employees are postgraduates?**
- A. 35% B. 85% C. 40% D. 50% E. None of these**
- 

13. X's expenditure is 25% more than Y's expenditure whereas Y's expenditure is 40% less than Z's expenditure. By what percentage is X's expenditure less than Z's expenditure?

A. 20% B. 35% C. 25% D. 30% E. None of these



14. Mr. Mirza Beg spends 80% of his income. If his income is increased by 20% and his expenditure is increased by 10% then what will be the percentage increase in his savings?

A. 15% B. 60% C. 30% D. 45% E. None of these



- 15. The price of an article has increased by 12%, 20% and 25% successively. What is the effective percentage increase in the price of the article?**
- A. 64% B. 68% C. 72% D. 76% E. None of these**
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16. The price of an article is increased by 15% and 20% successively and then decreased by 25%. What is the effective percentage change in the price?

A. 3.5% increase

B. 2.5% increase

C. 4.5% increase

D. 10% increase

E. NOT

17. When the price of an article is decreased by 40% its sales increase by 50%

What is the effect on the revenue of the article?

A. 10% increase

B. 10% decrease

C. 2% decrease

D. 2% increase

E. NOT

- 18. The length and the breadth of a rectangle are increased by 15% and 20% respectively. What is the percentage increase in the area of the rectangle?**
- A. 38% B. 42% C. 46% D. 50% E. None of these**

19. Mr. Rahul spends 20% of his income on house rent, 28% on food, 11% on clothes, 15% on education, 5% on travel, 8% on medicines and saves 1,820. What is the total expenditure on education and medicine?

A. ₹5,460

B. ₹2,240

C. ₹4,030

D. 3,320

E. None of these



20. Sumit spends 20% of her income on house rent, 10% of the remaining on travel, 30% of the remaining on clothes, $14\frac{2}{7}\%$ of the remaining on education, 50% of the remaining on food, $33\frac{1}{3}\%$ of the remaining on medicines and saves remaining amount of ₹2,304. What is the expenditure on education?

A. ₹1152

B. ₹6912

C. ₹4535

D. ₹1048

E. None of these

21. The population of a country in 1991 was 25 crores and in 2001 it was 12% more than what it was in 1991. While the population in 2011 was 15% more than what it was in 2001. What was the population of the country in 2011?

- A. 32.2 crores B. 34.6 crores C. 36.8 crores**
D. 38.4 crores E. None of these

22. In a colony of 800 members, 55% are females. Of the males, 35% are employed while 25% of the females are employed. What percentage of the total number of members of the colony are employed?

A. 29.5%

B. 34.5%

C. 39.5%

D. 44.5%

E. None of these



23. In an election, three candidates A, B and C contested. Of the votes polled. A got 42%, B got 28% of the votes polled and C got the rest. A got 3096 more votes than C. How many votes were polled in total ?

A. 21600

B. 22900

C. 24200

D. 25800 E. None of these

24. Anil invested 10% less than Bala and Bala invested 10% more than Charan. If the sum total of their investments is 12,360, then how much amount did Charan invest?

A. ₹4,020

B. ₹4,000

C. ₹4,200

D. ₹4,420 E. None of these

- 25. In city XYZ of population 12 lakhs, 70% of the population is aged above 18 years and of this 20% is above 60 years. Find the number of people aged above 18 years but aged 60 years and below.**
- A. 6 lakhs B. lakhs C. 6.96 lakhs D. 7.02 lakhs E. None of these**
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26. In an examination, 40% of the total number of students failed in English, 25% failed in Hindi, 10% failed in both the subjects. Find the percentage of those who passed in both the subjects.

A. 55%

B. 45%

C. 65%

D. 35%

E. None of these



27. In a village election, only two candidates contested and 80% of registered voters cast their votes. There are no invalid votes. The winner got 900 votes more than his opponent. If the loser had got 30% more votes, there would have been a tie between them. Find the number of registered voters in the village.

- A. 3900 B. 4875 C. 3785 D. 3875 E. None of these**
- 

28. If the present population of a state is 3,25 000 and after two years it increases to 4,41 ,025, then what is the rate of increase for the first year if it is known that the rate of increase for the second year is 18%?
A. 10% B. 12% C. 15% D. 17% E. None of these



29. The present population of a town is 42,000, and every year it increases at a constant rate. If after two years, it increases to 55,545, what is the rate of increase?
A. 7.5% B. 15% C. 30% D. 10% E. None of these



30. The salary of a person was increased by 20% at the end of the first year and then reduced by 15% at the end of the second year and as a result his salary is ₹ 17,595. What was his salary two years ago?

- A. ₹17,000 B. ₹17,250 C. ₹17,500**
D. 18,250 E. None of these

31. Fresh orange contains 95% water while dry orange contains 12% water. If the weight of a dry orange is 50 grams, what was its total weight when it was fresh ?
A. 440 g B. 550 g C. 880 g D. 980 g E. None of these



32. If the numerator of a fraction increases by 418% and the denominator increases by 117%, then the value of the fraction is $\frac{3}{31}$. What is the original fraction ?

A. $\frac{11}{82}$

B. $\frac{3}{74}$

C. $\frac{5}{76}$

D. $\frac{2}{83}$

33. A% of A is the same as 10% of

A. $\frac{A^3}{10}$

B. $A/100$

C. $\frac{A^2}{100}$

D. None

34. If $a\%$ of x is equal to $b\%$ of y , then $c\%$ of y is what percent of x ?
- A. $c\%$ B. $(ac/b)\%$ C. $(bc/a)\%$ D. $abc\%$

35. In a class of 52 students, 25% are rich and others are poor. There are 20 females in the class, of whom 55% are poor. How many rich males are there in the class.

- A. 13 B. 4 C. 39 D. 2 E. 28**
- 

36. If $\frac{1}{8}$ of $\frac{2}{3}$ of $\frac{4}{5}$ of a number is 12 then 30% of the number will be
A. 48 B. 64 C. 54 D. 42 E. None

37. A positive number is by mistake divided by 6 instead of being multiplied by 6. What is the % error based on correct answer?

A. 3 B. 97 C. 17 D. 83 E. None

38. What is the total number of customers in a shop on a particular day, when 31% of them purchase on credit, and the number of those who do cash purchase exceeds the number of credit purchases by 247 ?

- A. 605` B. 560 C. 650 D. 1650 E. None**

39. In an election between two candidates, 70% of the total votes enrolled were polled. Out of which 4% votes were declared invalid. The winning candidate got 12600 votes, which were 60% of the valid votes. The number of voters cast their vote will be

A. 31250

B. 32050

C. 21875

D. 22125



40. If 12% of x is equally to 6% of y , then 18% of x will be equal to how much % of y ?

A. 7% B. 9% C. 11% D. None

THANK YOU

